

SKILL WEEK-4 : ACID PROPERTIES AND ISOLATION

1. CREATE DATABASE

CREATE DATABASE Bank_ACID_DB;

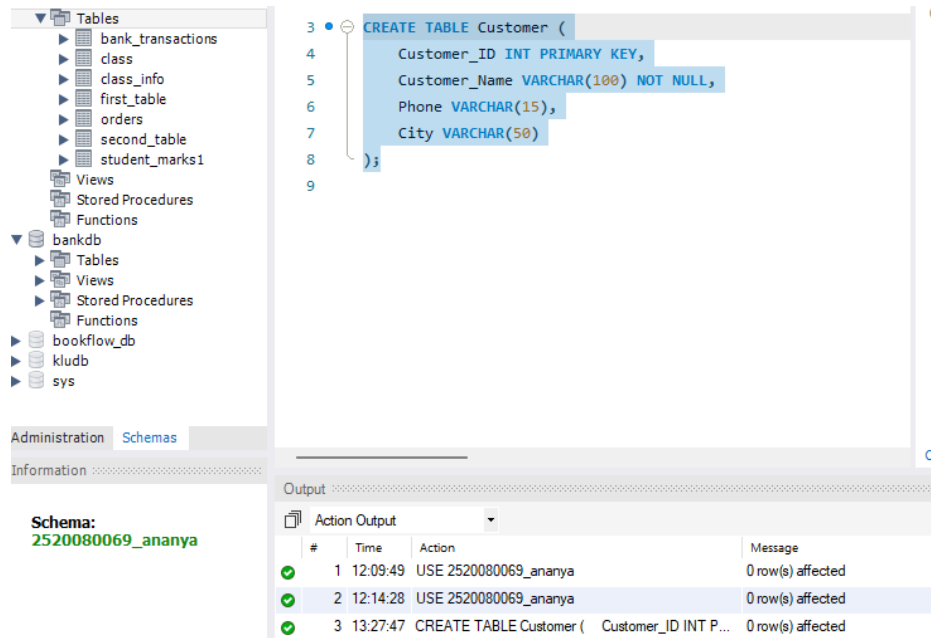
USE Bank_ACID_DB;

The screenshot shows the SQL Enterprise Manager interface. On the left, the 'SCHEMAS' tree is expanded, showing the '2520080069_ananya' database. The 'Tables' folder is expanded, showing a list of tables: bank_transactions, class, class_info, first_table, orders, second_table, and student_marks1. The 'Views' folder is also expanded, showing a list of views: bank_transactions, class, class_info, first_table, orders, second_table, and student_marks1. The 'Stored Procedures' and 'Functions' folders are also expanded. The 'Information' tab is selected, showing the 'Schema: 2520080069_ananya'. The 'Output' window at the bottom shows the execution of the 'USE 2520080069_ananya;' statement, with two rows of output: Row 1: 12:09:49, USE 2520080069_ananya, 0 row(s) affected; Row 2: 12:14:28, USE 2520080069_ananya, 0 row(s) affected.

#	Time	Action	Message
1	12:09:49	USE 2520080069_ananya	0 row(s) affected
2	12:14:28	USE 2520080069_ananya	0 row(s) affected

2. CREATE CUSTOMER TABLE

CREATE TABLE Customer (
 Customer_ID INT PRIMARY KEY,
 Customer_Name VARCHAR(100) NOT NULL,
 Phone VARCHAR(15),
 City VARCHAR(50)
);



3. CREATE ACCOUNT TABLE

```
CREATE TABLE Account (
  Account_No INT PRIMARY KEY,
  Customer_ID INT,
  Account_Type VARCHAR(20),
  Balance DECIMAL(12,2),
  Branch VARCHAR(50),
  FOREIGN KEY (Customer_ID)
  REFERENCES Customer(Customer_ID)
);
```

```
10 CREATE TABLE Account (  
11     Account_No INT PRIMARY KEY,  
12     Customer_ID INT,  
13     Account_Type VARCHAR(20),  
14     Balance DECIMAL(12,2),  
15     Branch VARCHAR(50),  
16     FOREIGN KEY (Customer_ID)  
17     REFERENCES Customer(Customer_ID)  
18 );  
19
```

Output

Action Output

#	Time	Action	Message
✓ 1	12:09:49	USE 2520080069_ananya	0 row(s) affected
✓ 2	12:14:28	USE 2520080069_ananya	0 row(s) affected
✓ 3	13:27:47	CREATE TABLE Customer (Customer_ID INT P...	0 row(s) affected
✓ 4	13:28:21	CREATE TABLE Account (Account_No INT PR...	0 row(s) affected

4. CREATE TRANSACTION TABLE

```
CREATE TABLE Bank_Transaction (  
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,  
    Account_No INT,  
    Transaction_Type VARCHAR(20),  
    Amount DECIMAL(12,2),  
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (Account_No)  
    REFERENCES Account(Account_No)  
);
```

```
20 CREATE TABLE Bank_Transaction (  
21     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,  
22     Account_No INT,  
23     Transaction_Type VARCHAR(20),  
24     Amount DECIMAL(12,2),  
25     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,  
26     FOREIGN KEY (Account_No)  
27     REFERENCES Account(Account_No)  
28 );  
29
```

Context

Output

Action Output

#	Time	Action	Message
✓ 1	12:09:49	USE 2520080069_ananya	0 row(s) affected
✓ 2	12:14:28	USE 2520080069_ananya	0 row(s) affected
✓ 3	13:27:47	CREATE TABLE Customer (Customer_ID INT P...	0 row(s) affected
✓ 4	13:28:21	CREATE TABLE Account (Account_No INT PR...	0 row(s) affected
✓ 5	13:28:56	CREATE TABLE Bank_Transaction (Transactio...	0 row(s) affected

5. INSERT CUSTOMER DATA

INSERT INTO Customer

(Customer_ID, Customer_Name, Phone, City)

VALUES

(101, 'Ravi Kumar', '9876543210', 'Hyderabad'),
(102, 'Priya Sharma', '9876543211', 'Vijayawada'),
(103, 'Arjun Reddy', '9876543212', 'Bangalore'),
(104, 'Sneha Rao', '9876543213', 'Chennai'),
(105, 'Kiran Kumar', '9876543214', 'Hyderabad');

SCHEMAS

Filter objects

2520080069_ananya

- Tables
 - bank_transactions
 - class
 - class_info
 - first_table
 - orders
 - second_table
 - student_marks1
- Views
- Stored Procedures
- Functions

bankdb

- Tables
- Views
- Stored Procedures
- Functions

bookflow_db

kludb

sys

Administration Schemas

Information

Schema: 2520080069_ananya

```

22 Account_No INT,
23 Transaction_Type VARCHAR(20),
24 Amount DECIMAL(12,2),
25 Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
26 FOREIGN KEY (Account_No)
27 REFERENCES Account(Account_No)
28 );
29
30 INSERT INTO Customer
31 (Customer_ID, Customer_Name, Phone, City)
32 VALUES
33 (101, 'Ravi Kumar', '9876543210', 'Hyderabad'),
34 (102, 'Priya Sharma', '9876543211', 'Vijayawada'),
35 (103, 'Arjun Reddy', '9876543212', 'Bangalore'),
36 (104, 'Sneha Rao', '9876543213', 'Chennai'),
37 (105, 'Kiran Kumar', '9876543214', 'Hyderabad');
38
39

```

Automatic commit disabled. Use the manually get current caret position to toggle automatic commit.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	
1	12:09:49	USE 2520080069_ananya	0 row(s) affected	0
2	12:14:28	USE 2520080069_ananya	0 row(s) affected	0
3	13:27:47	CREATE TABLE Customer (Customer_ID INT P...	0 row(s) affected	0
4	13:28:21	CREATE TABLE Account (Account_No INT PR...	0 row(s) affected	0
5	13:28:56	CREATE TABLE Bank_Transaction (Transactio...	0 row(s) affected	0
6	13:29:41	INSERT INTO Customer (Customer_ID, Customer_...	5 row(s) affected Records: 5 Duplicates: 0 Warnin...	0

6. INSERT ACCOUNT DATA

INSERT INTO Account

(Account_No, Customer_ID, Account_Type, Balance, Branch)

VALUES

(10001, 101, 'Savings', 50000, 'Hyderabad'),

(10002, 102, 'Savings', 75000, 'Vijayawada'),

(10003, 103, 'Current', 120000, 'Bangalore'),

(10004, 104, 'Savings', 45000, 'Chennai'),

(10005, 105, 'Current', 90000, 'Hyderabad');

```

39 • INSERT INTO Account
40 (Account_No, Customer_ID, Account_Type, Balance, Branch)
41 VALUES
42 (10001, 101, 'Savings', 50000, 'Hyderabad'),
43 (10002, 102, 'Savings', 75000, 'Vijayawada'),
44 (10003, 103, 'Current', 120000, 'Bangalore'),
45 (10004, 104, 'Savings', 45000, 'Chennai'),
46 (10005, 105, 'Current', 90000, 'Hyderabad');
47

```

Context Help Snipp

Output

Action Output

#	Time	Action	Message
✓ 1	12:09:49	USE 2520080069_ananya	0 row(s) affected
✓ 2	12:14:28	USE 2520080069_ananya	0 row(s) affected
✓ 3	13:27:47	CREATE TABLE Customer (Customer_ID INT P...	0 row(s) affected
✓ 4	13:28:21	CREATE TABLE Account (Account_No INT PR...	0 row(s) affected
✓ 5	13:28:56	CREATE TABLE Bank_Transaction (Transactio...	0 row(s) affected
✓ 6	13:29:41	INSERT INTO Customer (Customer_ID, Customer_...	5 row(s) affected Records: 5 Duplicates: 0 Wamin...
✓ 7	13:30:28	INSERT INTO Account (Account_No, Customer_ID...	5 row(s) affected Records: 5 Duplicates: 0 Wamin...

7. INSERT BANK TRANSACTION DATA

```

INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000);

```

```

48 • INSERT INTO Bank_Transaction
49 (Account_No, Transaction_Type, Amount)
50 VALUES
51 (10001, 'DEPOSIT', 10000),
52 (10002, 'DEPOSIT', 15000),
53 (10003, 'WITHDRAW', 20000),
54 (10004, 'DEPOSIT', 5000),
55 (10005, 'WITHDRAW', 10000);
56

```

Context Help Snipp

Output

Action Output

#	Time	Action	Message
✓ 1	12:09:49	USE 2520080069_ananya	0 row(s) affected
✓ 2	12:14:28	USE 2520080069_ananya	0 row(s) affected
✓ 3	13:27:47	CREATE TABLE Customer (Customer_ID INT P...	0 row(s) affected
✓ 4	13:28:21	CREATE TABLE Account (Account_No INT PR...	0 row(s) affected
✓ 5	13:28:56	CREATE TABLE Bank_Transaction (Transactio...	0 row(s) affected
✓ 6	13:29:41	INSERT INTO Customer (Customer_ID, Customer_...	5 row(s) affected Records: 5 Duplicates: 0 Wamin...
✓ 7	13:30:28	INSERT INTO Account (Account_No, Customer_ID...	5 row(s) affected Records: 5 Duplicates: 0 Wamin...
✓ 8	13:30:57	INSERT INTO Bank_Transaction (Account_No, Tr...	5 row(s) affected Records: 5 Duplicates: 0 Wamin...

8. DISPLAY DATA

SELECT * FROM Customer;

bookflow_db
kludb
sys

```

57 • SELECT * FROM Customer;
58 • SELECT * FROM Account;
59 • SELECT * FROM Bank_Transaction;

```

Administration Schemas

Information

Schema: 2520080069_ananya

Result Grid

Customer_ID	Customer_Name	Phone	City
101	Ravi Kumar	9876543210	Hyderabad
102	Priya Sharma	9876543211	Vijayawada
103	Arjun Reddy	9876543212	Bangalore
104	Sneha Rao	9876543213	Chennai
105	Kiran Kumar	9876543214	Hyderabad
NULL	NULL	NULL	NULL

SELECT * FROM Account;

bookflow_db
kludb
sys

```

57 • SELECT * FROM Customer;
58 • SELECT * FROM Account;
59 • SELECT * FROM Bank_Transaction;

```

Administration Schemas

Information

Schema: 2520080069_ananya

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Bank_Transaction;
```

The screenshot shows a database management tool interface. On the left, there's a sidebar with a tree view containing 'bookflow_db', 'kludb', and 'sys'. Below this, the 'Administration' tab is active, showing 'Schemas' and 'Information'. The 'Schema' is listed as '2520080069_ananya'. The main window displays a query window with three lines of SQL code: '57 • SELECT * FROM Customer;', '58 • SELECT * FROM Account;', and '59 • SELECT * FROM Bank_Transaction;'. Below the query window, the 'Result Grid' is visible, showing a table with 6 columns: 'Transaction_ID', 'Account_No', 'Transaction_Type', 'Amount', and 'Transaction_Date'. The table contains 5 rows of data, with the last row showing 'NULL' values for all columns.

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-14 13:30:57
2	10002	DEPOSIT	15000.00	2026-08-14 13:30:57
3	10003	WITHDRAW	20000.00	2026-08-14 13:30:57
4	10004	DEPOSIT	5000.00	2026-08-14 13:30:57
5	10005	WITHDRAW	10000.00	2026-08-14 13:30:57
NULL	NULL	NULL	NULL	NULL

PART A – TRANSACTIONS

9. START A TRANSACTION

```
START TRANSACTION;
```

```
61 • START TRANSACTION;  
62
```

The screenshot shows a database management tool interface. The 'Output' tab is active, displaying a list of actions and messages. The table has 4 columns: '#', 'Time', 'Action', and 'Message'. The actions include 'INSERT INTO Account', 'INSERT INTO Bank_Transaction', and 'SELECT * FROM' queries, along with a 'START TRANSACTION' action.

#	Time	Action	Message
7	13:30:28	INSERT INTO Account (Account_No, Customer_ID,...	5 row(s) affected Records
8	13:30:57	INSERT INTO Bank_Transaction (Account_No, Tra...	5 row(s) affected Records
9	13:31:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	13:31:24	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	13:31:24	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	13:32:52	START TRANSACTION	0 row(s) affected

10. DEPOSIT MONEY

```
UPDATE Account
```

```
SET Balance = Balance + 5000
```

```
WHERE Account_No = 10001;
```



```

60
61 • START TRANSACTION;
62 • UPDATE Account
63   SET Balance = Balance + 5000
64   WHERE Account_No = 10001;

```

Context Help Snipp

Output

Action Output

#	Time	Action	Message	Du
6	13:29:41	INSERT INTO Customer (Customer_ID, Customer_N...	5 row(s) affected Records: 5 Duplicates: 0 Warning...	0.0
7	13:30:28	INSERT INTO Account (Account_No, Customer_ID,...	5 row(s) affected Records: 5 Duplicates: 0 Warning...	0.0
8	13:30:57	INSERT INTO Bank_Transaction (Account_No, Tra...	5 row(s) affected Records: 5 Duplicates: 0 Warning...	0.0
9	13:31:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.0
10	13:31:24	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.0
11	13:31:24	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.0
12	13:32:52	START TRANSACTION	0 row(s) affected	0.0
13	13:33:15	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...	0.0

11. CHECK BALANCE

```

SELECT *
FROM Account
WHERE Account_No = 10001;

```

Stored Procedures
Functions
bookflow_db
kludb
sys

```

67 • SELECT *
68 FROM Account
69 WHERE Account_No = 10001;
70

```

Administration Schemas

Information

Schema:
2520080069_ananya

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

12. COMMIT TRANSACTION

```

COMMIT;

```

```

71 • COMMIT;

```

Output

Action Output

#	Time	Action	Message
8	13:30:57	INSERT INTO Bank_Transaction (Account_No, Tra...	5 row(s) affected Records: 5 Duplicates...
9	13:31:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	13:31:24	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	13:31:24	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	13:32:52	START TRANSACTION	0 row(s) affected
13	13:33:15	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Cha...
14	13:33:51	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
15	13:34:17	COMMIT	0 row(s) affected

13. CHECK BALANCE AFTER COMMIT

```
SELECT *  
FROM Account  
WHERE Account_No = 10001;
```

The screenshot shows a database management tool interface. On the left, there is a tree view with folders for Views, Stored Procedures, Functions, bookflow_db, kludb, and sys. The 'Schemas' tab is selected, showing a schema named '2520080069_ananya'. The main area displays a SQL query in a text editor:

```
73 • SELECT *  
74 FROM Account  
75 WHERE Account_No = 10001;  
76
```

Below the query editor, there is a 'Result Grid' tab. The grid shows the following data:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

At the bottom, there are buttons for 'Administration', 'Schemas', 'Information', 'Filter Rows:', 'Edit:', and 'Export/Import:'.

PART B – ROLLBACK

14. START TRANSACTION

```
START TRANSACTION;
```

The screenshot shows a database management tool interface. The top section displays a SQL query in a text editor:

```
76  
77 • START TRANSACTION;
```

Below the query editor, there is an 'Output' tab. The output shows a list of actions and their results:

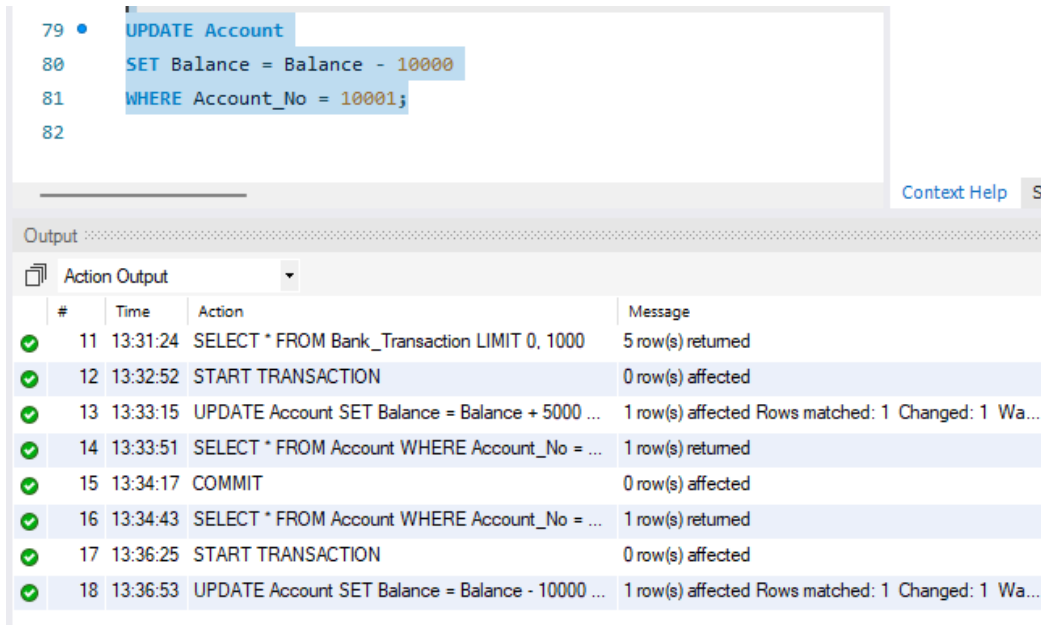
#	Time	Action	Message
✓ 10	13:31:24	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 11	13:31:24	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
✓ 12	13:32:52	START TRANSACTION	0 row(s) affected
✓ 13	13:33:15	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Changed: 1 W
✓ 14	13:33:51	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 15	13:34:17	COMMIT	0 row(s) affected
✓ 16	13:34:43	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 17	13:36:25	START TRANSACTION	0 row(s) affected

At the bottom right, there is a 'Context Help' button.

15. WITHDRAW MONEY

```
UPDATE Account  
SET Balance = Balance - 10000
```

WHERE Account_No = 10001;



The screenshot shows a SQL IDE interface. The script editor at the top contains the following SQL code:

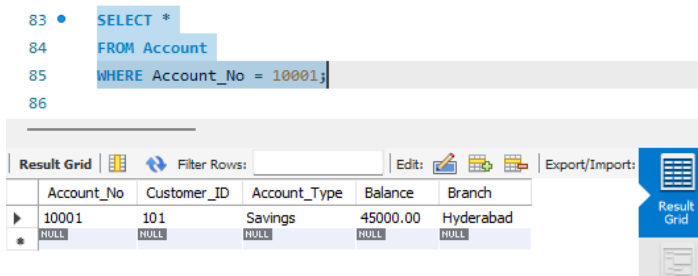
```
79 • UPDATE Account
80 SET Balance = Balance - 10000
81 WHERE Account_No = 10001;
82
```

Below the script editor is the "Output" window, which is currently displaying the "Action Output". It shows a list of database actions and their results:

#	Time	Action	Message
✓ 11	13:31:24	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
✓ 12	13:32:52	START TRANSACTION	0 row(s) affected
✓ 13	13:33:15	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...
✓ 14	13:33:51	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 15	13:34:17	COMMIT	0 row(s) affected
✓ 16	13:34:43	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 17	13:36:25	START TRANSACTION	0 row(s) affected
✓ 18	13:36:53	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...

16. CHECK BALANCE

SELECT *
FROM Account
WHERE Account_No = 10001;



The screenshot shows a SQL IDE interface. The script editor at the top contains the following SQL code:

```
83 • SELECT *
84 FROM Account
85 WHERE Account_No = 10001;
86
```

Below the script editor is the "Result Grid" window, which displays the results of the query:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad
* NULL	NULL	NULL	NULL	NULL

17. ROLLBACK

ROLLBACK;

87	•	ROLLBACK;
88		

#	Time	Action	Message
✓ 13	13:33:15	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1
✓ 14	13:33:51	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 15	13:34:17	COMMIT	0 row(s) affected
✓ 16	13:34:43	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 17	13:36:25	START TRANSACTION	0 row(s) affected
✓ 18	13:36:53	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1
✓ 19	13:37:18	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 20	13:38:25	ROLLBACK	0 row(s) affected

18. CHECK BALANCE AFTER ROLLBACK

```
SELECT *
FROM Account
WHERE Account_No = 10001;
```

89	•	SELECT *
90		FROM Account
91		WHERE Account_No = 10001;
92		

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad
*	NULL	NULL	NULL	NULL

PART C – SAVEPOINT

19. START TRANSACTION

```
START TRANSACTION;
```

93	•	START TRANSACTION;
----	---	--------------------

#	Time	Action	Message
✓ 15	13:34:17	COMMIT	0 row(s) affected
✓ 16	13:34:43	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 17	13:36:25	START TRANSACTION	0 row(s) affected
✓ 18	13:36:53	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...
✓ 19	13:37:18	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 20	13:38:25	ROLLBACK	0 row(s) affected
✓ 21	13:38:57	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 22	13:39:25	START TRANSACTION	0 row(s) affected

20. FIRST UPDATE

UPDATE Account
 SET Balance = Balance + 5000
 WHERE Account_No = 10001;

Schema: 2520080069_ananya	92	
	93	• START TRANSACTION;
	94	
	95	• UPDATE Account
	96	SET Balance = Balance + 5000
	97	WHERE Account_No = 10001;
	98	

#	Time	Action	Message
✓ 16	13:34:43	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 17	13:36:25	START TRANSACTION	0 row(s) affected
✓ 18	13:36:53	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...
✓ 19	13:37:18	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 20	13:38:25	ROLLBACK	0 row(s) affected
✓ 21	13:38:57	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 22	13:39:25	START TRANSACTION	0 row(s) affected
✓ 23	13:39:48	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...

21. CREATE SAVEPOINT

SAVEPOINT Deposit1;

99 • **SAVEPOINT Deposit1;**

Output

Action Output

#	Time	Action	Message
✓ 17	13:36:25	START TRANSACTION	0 row(s) affected
✓ 18	13:36:53	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1
✓ 19	13:37:18	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 20	13:38:25	ROLLBACK	0 row(s) affected
✓ 21	13:38:57	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 22	13:39:25	START TRANSACTION	0 row(s) affected
✓ 23	13:39:48	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1
✓ 24	13:40:20	SAVEPOINT Deposit1	0 row(s) affected

22. SECOND UPDATE

UPDATE Account

SET Balance = Balance - 3000

WHERE Account_No = 10002;

```
101 • UPDATE Account
102     SET Balance = Balance - 3000
103     WHERE Account_No = 10002;
104
```

Output

Action Output

#	Time	Action	Message
✓ 18	13:36:53	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1
✓ 19	13:37:18	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 20	13:38:25	ROLLBACK	0 row(s) affected
✓ 21	13:38:57	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 22	13:39:25	START TRANSACTION	0 row(s) affected
✓ 23	13:39:48	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1
✓ 24	13:40:20	SAVEPOINT Deposit1	0 row(s) affected
✓ 25	13:40:53	UPDATE Account SET Balance = Balance - 3000 ...	1 row(s) affected Rows matched: 1

23. CREATE SECOND SAVEPOINT

SAVEPOINT Withdrawal1;

```

105 • SAVEPOINT Withdrawal1;
106

```

Output

Action Output

#	Time	Action	Message
✓ 19	13:37:18	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 20	13:38:25	ROLLBACK	0 row(s) affected
✓ 21	13:38:57	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 22	13:39:25	START TRANSACTION	0 row(s) affected
✓ 23	13:39:48	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched:
✓ 24	13:40:20	SAVEPOINT Deposit1	0 row(s) affected
✓ 25	13:40:53	UPDATE Account SET Balance = Balance - 3000 ...	1 row(s) affected Rows matched:
✓ 26	13:41:15	SAVEPOINT Withdrawal1	0 row(s) affected

24. THIRD UPDATE

UPDATE Account

SET Balance = Balance + 10000

WHERE Account_No = 10003;

```

107 • UPDATE Account
108     SET Balance = Balance + 10000
109     WHERE Account_No = 10003;
110

```

Cor

Output

Action Output

#	Time	Action	Message
✓ 20	13:38:25	ROLLBACK	0 row(s) affected
✓ 21	13:38:57	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 22	13:39:25	START TRANSACTION	0 row(s) affected
✓ 23	13:39:48	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Cha
✓ 24	13:40:20	SAVEPOINT Deposit1	0 row(s) affected
✓ 25	13:40:53	UPDATE Account SET Balance = Balance - 3000 ...	1 row(s) affected Rows matched: 1 Cha
✓ 26	13:41:15	SAVEPOINT Withdrawal1	0 row(s) affected
✓ 27	13:41:40	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Cha

25. ROLLBACK TO SAVEPOINT

ROLLBACK TO SAVEPOINT Withdrawal1;

111 • `ROLLBACK TO SAVEPOINT Withdrawal1;`

Output

Action Output

#	Time	Action	Message
✓ 21	13:38:57	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 22	13:39:25	START TRANSACTION	0 row(s) affected
✓ 23	13:39:48	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1
✓ 24	13:40:20	SAVEPOINT Deposit1	0 row(s) affected
✓ 25	13:40:53	UPDATE Account SET Balance = Balance - 3000 ...	1 row(s) affected Rows matched: 1
✓ 26	13:41:15	SAVEPOINT Withdrawal1	0 row(s) affected
✓ 27	13:41:40	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1
✓ 28	13:42:04	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected

26. COMMIT

COMMIT;

113 • `COMMIT;`

Output

Action Output

#	Time	Action	Message
✓ 22	13:39:25	START TRANSACTION	0 row(s) affected
✓ 23	13:39:48	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1
✓ 24	13:40:20	SAVEPOINT Deposit1	0 row(s) affected
✓ 25	13:40:53	UPDATE Account SET Balance = Balance - 3000 ...	1 row(s) affected Rows matched: 1
✓ 26	13:41:15	SAVEPOINT Withdrawal1	0 row(s) affected
✓ 27	13:41:40	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1
✓ 28	13:42:04	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
✓ 29	13:42:27	COMMIT	0 row(s) affected

27. CHECK ACCOUNTS

SELECT *
FROM Account;

115 • **SELECT ***
 116 **FROM Account;**
 117

Result Grid | Filter Rows: | Edit: | Export/Import:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	60000.00	Hyderabad
	10002	102	Savings	72000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Schema: 2520080069_ananya

PART D – BANK TRANSFER

28. TRANSFER MONEY

START TRANSACTION;

118 • **START TRANSACTION;**
 119

Context Help

Output

Action Output

#	Time	Action	Message
✓ 24	13:40:20	SAVEPOINT Deposit1	0 row(s) affected
✓ 25	13:40:53	UPDATE Account SET Balance = Balance - 3000 ...	1 row(s) affected Rows matched: 1 Changed: 1
✓ 26	13:41:15	SAVEPOINT Withdrawal1	0 row(s) affected
✓ 27	13:41:40	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Changed: 1
✓ 28	13:42:04	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
✓ 29	13:42:27	COMMIT	0 row(s) affected
✓ 30	13:42:51	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 31	13:44:24	START TRANSACTION	0 row(s) affected

29. DEDUCT FROM SENDER

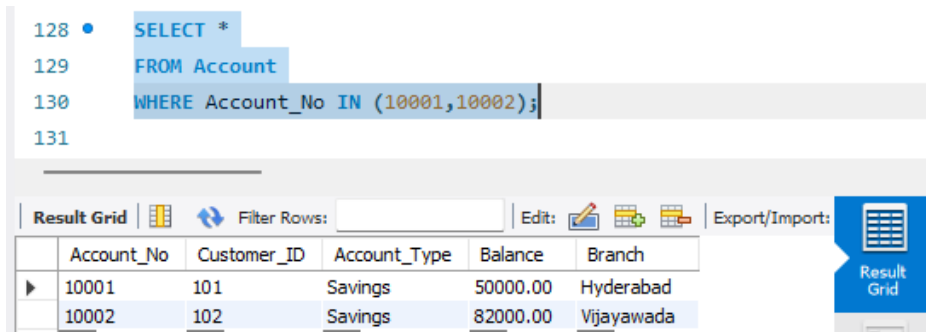
UPDATE Account

SET Balance = Balance - 10000

WHERE Account_No = 10001;

31. CHECK BOTH ACCOUNTS

```
SELECT *  
FROM Account  
WHERE Account_No IN (10001,10002);
```



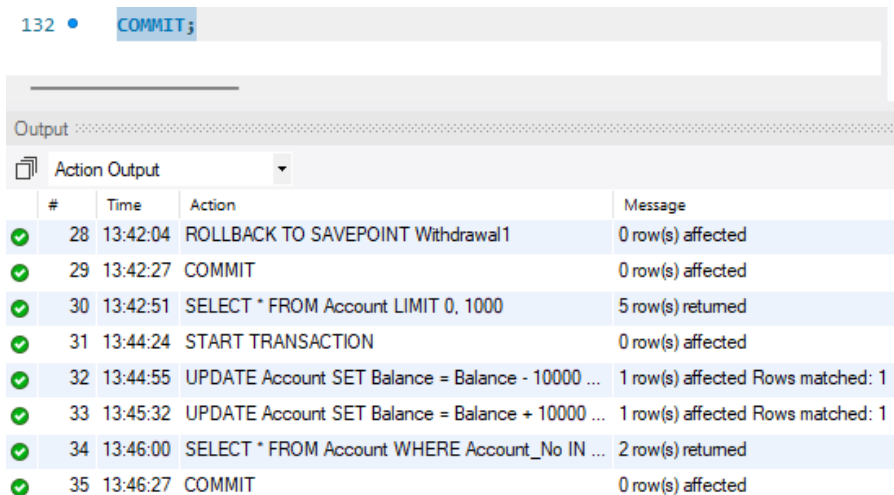
128 • SELECT *
129 FROM Account
130 WHERE Account_No IN (10001,10002);
131

Result Grid | Filter Rows: | Edit: | Export/Import: | Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	82000.00	Vijayawada

32. COMMIT TRANSFER

```
COMMIT;
```



132 • COMMIT;

Output :

Action Output

#	Time	Action	Message
✓ 28	13:42:04	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
✓ 29	13:42:27	COMMIT	0 row(s) affected
✓ 30	13:42:51	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 31	13:44:24	START TRANSACTION	0 row(s) affected
✓ 32	13:44:55	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1
✓ 33	13:45:32	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1
✓ 34	13:46:00	SELECT * FROM Account WHERE Account_No IN ...	2 row(s) returned
✓ 35	13:46:27	COMMIT	0 row(s) affected

33. FINAL BALANCES

```
SELECT  
    Account_No,  
    Balance  
FROM Account  
WHERE Account_No IN (10001,10002);
```

The screenshot shows a database management interface. On the left, a tree view displays the database structure: Functions, bankdb (Tables, Views, Stored Procedures, Functions), bookflow_db, kludb, and sys. The main area shows a SQL query being executed:

```

134 • SELECT
135     Account_No,
136     Balance
137 FROM Account
138 WHERE Account_No IN (10001,10002);
139

```

Below the query, the 'Result Grid' is displayed with the following data:

Account_No	Balance
10001	50000.00
10002	82000.00
NULL	NULL

The interface also includes tabs for 'Administration' and 'Schemas', and a 'Schema: 2520080069_ananya' label.

PART E – TRANSFER USING ROLLBACK

34. START TRANSACTION

START TRANSACTION;

The screenshot shows a list of SQL actions and their results. The 'Action Output' tab is selected, displaying a table with the following data:

#	Time	Action	Message
30	13:42:51	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
31	13:44:24	START TRANSACTION	0 row(s) affected
32	13:44:55	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Changed
33	13:45:32	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Changed
34	13:46:00	SELECT * FROM Account WHERE Account_No IN ...	2 row(s) returned
35	13:46:27	COMMIT	0 row(s) affected
36	13:46:50	SELECT Account_No, Balance FROM Account...	2 row(s) returned
37	13:49:16	START TRANSACTION	0 row(s) affected

35. DEDUCT MONEY

UPDATE Account
 SET Balance = Balance - 20000
 WHERE Account_No = 10001;

```

142 • UPDATE Account
143 SET Balance = Balance - 20000
144 WHERE Account_No = 10001;
145

```

Output

Action Output

#	Time	Action	Message
✓ 31	13:44:24	START TRANSACTION	0 row(s) affected
✓ 32	13:44:55	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Change
✓ 33	13:45:32	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Change
✓ 34	13:46:00	SELECT * FROM Account WHERE Account_No IN ...	2 row(s) returned
✓ 35	13:46:27	COMMIT	0 row(s) affected
✓ 36	13:46:50	SELECT Account_No, Balance FROM Account...	2 row(s) returned
✓ 37	13:49:16	START TRANSACTION	0 row(s) affected
✓ 38	13:49:42	UPDATE Account SET Balance = Balance - 20000 ...	1 row(s) affected Rows matched: 1 Change

36. ADD MONEY

UPDATE Account
 SET Balance = Balance + 20000
 WHERE Account_No = 10002;

```

146 • UPDATE Account
147 SET Balance = Balance + 20000
148 WHERE Account_No = 10002;
149

```

Output

Action Output

#	Time	Action	Message
✓ 32	13:44:55	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Change
✓ 33	13:45:32	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Change
✓ 34	13:46:00	SELECT * FROM Account WHERE Account_No IN ...	2 row(s) returned
✓ 35	13:46:27	COMMIT	0 row(s) affected
✓ 36	13:46:50	SELECT Account_No, Balance FROM Account...	2 row(s) returned
✓ 37	13:49:16	START TRANSACTION	0 row(s) affected
✓ 38	13:49:42	UPDATE Account SET Balance = Balance - 20000 ...	1 row(s) affected Rows matched: 1 Change
✓ 39	13:50:04	UPDATE Account SET Balance = Balance + 20000 ...	1 row(s) affected Rows matched: 1 Change

37. CANCEL TRANSACTION

ROLLBACK;

150 • `ROLLBACK;`

Output

Action Output

#	Time	Action	Message
✓ 33	13:45:32	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Ch
✓ 34	13:46:00	SELECT * FROM Account WHERE Account_No IN ...	2 row(s) returned
✓ 35	13:46:27	COMMIT	0 row(s) affected
✓ 36	13:46:50	SELECT Account_No, Balance FROM Accoun...	2 row(s) returned
✓ 37	13:49:16	START TRANSACTION	0 row(s) affected
✓ 38	13:49:42	UPDATE Account SET Balance = Balance - 20000 ...	1 row(s) affected Rows matched: 1 Ch
✓ 39	13:50:04	UPDATE Account SET Balance = Balance + 20000 ...	1 row(s) affected Rows matched: 1 Ch
✓ 40	13:50:26	ROLLBACK	0 row(s) affected

38. CHECK BALANCES

SELECT *
FROM Account
WHERE Account_No IN (10001,10002);

Views
Stored Procedures
Functions
bookflow_db
kludb
sys

152 • `SELECT *`
153 `FROM Account`
154 `WHERE Account_No IN (10001,10002);`
155

Administration Schemas

Information

Schema:
2520080069_ananya

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	82000.00	Vijayawada
NULL	NULL	NULL	NULL	NULL

Result Grid
Form Editor

PART F – ACID

39. ATOMICITY

START TRANSACTION;
UPDATE Account

```

SET Balance = Balance - 10000
WHERE Account_No = 10001;
UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10002;
COMMIT;

```

The screenshot shows a database management tool interface. On the left, there's a sidebar with 'Administration' and 'Schemas' tabs. The 'Schemas' tab is active, showing a schema named '2520080069_ananya'. The main area displays a SQL script with line numbers 156 to 165. The script contains the following SQL statements:

```

156 • START TRANSACTION;
157 • UPDATE Account
158   SET Balance = Balance - 10000
159   WHERE Account_No = 10001;
160 • UPDATE Account
161   SET Balance = Balance + 10000
162   WHERE Account_No = 10002;
163 • COMMIT;
164
165

```

Below the SQL editor is an 'Output' window. It has a dropdown menu set to 'Action Output'. The output table shows the following rows:

#	Time	Action	Message
38	13:49:42	UPDATE Account SET Balance = Balance - 20000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...
39	13:50:04	UPDATE Account SET Balance = Balance + 20000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...
40	13:50:26	ROLLBACK	0 row(s) affected
41	13:50:55	SELECT * FROM Account WHERE Account_No IN ...	2 row(s) returned
42	13:51:32	START TRANSACTION	0 row(s) affected
43	13:51:37	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...
44	13:51:42	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...
45	13:51:45	COMMIT	0 row(s) affected

40. CONSISTENCY

Before transfer:

Account 10001 = 50000

Account 10002 = 75000

Total = 125000

After transferring 10000:

Account 10001 = 40000

Account 10002 = 85000

Total = 125000

The total money remains consistent.

41. ISOLATION

Isolation means:

Multiple transactions executing at the same time should not incorrectly interfere with each other.

Example:

Transaction T1:

Transfer money.

Transaction T2:

Read account balance.

The database controls how much of T1 is visible to T2.

42. DURABILITY

START TRANSACTION;

UPDATE Account

SET Balance = Balance + 5000

WHERE Account_No = 10001;

COMMIT;

The screenshot shows a database management tool interface. On the left, the 'Schema' is set to '2520080069_ananya'. The main area displays a SQL script with the following lines:

```
164  
165 • START TRANSACTION;  
166 • UPDATE Account  
167 SET Balance = Balance + 5000  
168 WHERE Account_No = 10001;  
169 • COMMIT;  
170
```

Below the script, the 'Output' tab is selected, showing a table of execution results:

#	Time	Action	Message
41	13:50:55	SELECT * FROM Account WHERE Account_No IN ...	2 row(s) returned
42	13:51:32	START TRANSACTION	0 row(s) affected
43	13:51:37	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1
44	13:51:42	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1
45	13:51:45	COMMIT	0 row(s) affected
46	13:52:23	START TRANSACTION	0 row(s) affected
47	13:52:23	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1
48	13:52:23	COMMIT	0 row(s) affected

At the bottom left, there are tabs for 'Object Info' and 'Session'.

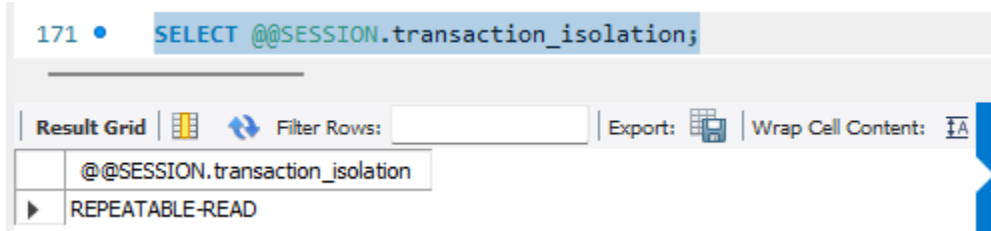
PART G – TRANSACTION ISOLATION LEVELS

43. CHECK CURRENT ISOLATION LEVEL

SELECT @@SESSION.transaction_isolation;

OR:

SELECT @@GLOBAL.transaction_isolation;



44. READ UNCOMMITTED

SET SESSION TRANSACTION ISOLATION LEVEL
READ UNCOMMITTED;

START TRANSACTION;
SELECT *
FROM Account
WHERE Account_No = 10001;
COMMIT;

Views

Stored Procedures

Functions

bankdb

Tables

Views

Stored Procedures

Functions

bookflow_db

kludb

sys

Administration

Schemas

Schema:

2520080069_ananya

```

173 • SET SESSION TRANSACTION ISOLATION LEVEL
174 READ UNCOMMITTED;
175 • START TRANSACTION;
176 • SELECT *
177 FROM Account
178 WHERE Account_No = 10001;
179 • COMMIT;
180

```

Result Grid

Filter Rows:

Edit

Export/Import

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Field Types

Account 14

Apply

Revert

Output

Action Output

#	Time	Action	Message
✓ 46	13:52:23	START TRANSACTION	0 row(s) affected
✓ 47	13:52:23	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1
✓ 48	13:52:23	COMMIT	0 row(s) affected
✓ 49	13:52:52	SELECT @@SESSION.transaction_isolation LIMIT ...	1 row(s) returned
✓ 50	13:53:18	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 51	13:53:31	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 52	13:53:50	COMMIT	0 row(s) affected
✓ 53	13:54:11	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned

Object Info

Session

45. READ COMMITTED

```

SET SESSION TRANSACTION ISOLATION LEVEL
READ COMMITTED;
START TRANSACTION;
SELECT *
FROM Account
WHERE Account_No = 10001;
COMMIT;

```

```

181 • SET SESSION TRANSACTION ISOLATION LEVEL
182     READ COMMITTED;
183 • START TRANSACTION;
184 • SELECT *
185     FROM Account
186     WHERE Account_No = 10001;
187 • COMMIT;

```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad
* NULL	NULL	NULL	NULL	NULL

Account 15 x Apply Revert Co

#	Time	Action	Message
✓ 50	13:53:18	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 51	13:53:31	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 52	13:53:50	COMMIT	0 row(s) affected
✓ 53	13:54:11	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 54	13:54:49	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 55	13:54:49	START TRANSACTION	0 row(s) affected
✓ 56	13:54:49	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 57	13:54:49	COMMIT	0 row(s) affected

46. REPEATABLE READ

```

SET SESSION TRANSACTION ISOLATION LEVEL
REPEATABLE READ;
START TRANSACTION;
SELECT *
FROM Account
WHERE Account_No = 10001;
COMMIT;

```

189 • SET SESSION TRANSACTION ISOLATION LEVEL
 190 REPEATABLE READ;
 191 • START TRANSACTION;
 192 • SELECT *
 193 FROM Account
 194 WHERE Account_No = 10001;
 195 • COMMIT;

Result Grid | Filter Rows: | Edit: | Export/Import: | **Result Grid**

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Account 16 x Apply Revert

Output

Action Output

#	Time	Action	Message
✓ 54	13:54:49	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 55	13:54:49	START TRANSACTION	0 row(s) affected
✓ 56	13:54:49	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 57	13:54:49	COMMIT	0 row(s) affected
✓ 58	13:55:19	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 59	13:55:19	START TRANSACTION	0 row(s) affected
✓ 60	13:55:19	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 61	13:55:19	COMMIT	0 row(s) affected

47. SERIALIZABLE

```

SET SESSION TRANSACTION ISOLATION LEVEL
SERIALIZABLE;
START TRANSACTION;
SELECT *
FROM Account
WHERE Account_No = 10001;
COMMIT;

```

```

197 • SET SESSION TRANSACTION ISOLATION LEVEL
198     SERIALIZABLE;
199 • START TRANSACTION;
200 • SELECT *
201     FROM Account
202     WHERE Account_No = 10001;
203 • COMMIT;

```

Result Grid

Filter Rows:

Edit:

Export/Import:

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Field Types

Account 17 ×

Apply

Revert

Col

Output

Action Output

	#	Time	Action	Message
✓	58	13:55:19	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓	59	13:55:19	START TRANSACTION	0 row(s) affected
✓	60	13:55:19	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓	61	13:55:19	COMMIT	0 row(s) affected
✓	62	13:55:46	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓	63	13:55:46	START TRANSACTION	0 row(s) affected
✓	64	13:55:46	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓	65	13:55:46	COMMIT	0 row(s) affected

PART H – DIRTY READ EXPERIMENT

SESSION 1:

SET SESSION TRANSACTION ISOLATION LEVEL

READ UNCOMMITTED;

START TRANSACTION;

UPDATE Account

SET Balance = Balance + 20000

WHERE Account_No = 10001;

Schema:
2520080069_ananya

```
197 • SET SESSION TRANSACTION ISOLATION LEVEL
198   SERIALIZABLE;
199 • START TRANSACTION;
200 • SELECT *
201   FROM Account
202   WHERE Account_No = 10001;
203 • COMMIT;
```

Output

Action Output

#	Time	Action	Message
✓ 61	13:55:19	COMMIT	0 row(s) affected
✓ 62	13:55:46	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 63	13:55:46	START TRANSACTION	0 row(s) affected
✓ 64	13:55:46	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 65	13:55:46	COMMIT	0 row(s) affected
✓ 66	13:56:23	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 67	13:56:23	START TRANSACTION	0 row(s) affected
✓ 68	13:56:23	UPDATE Account SET Balance = Balance + 20000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...

SESSION 2:

SET SESSION TRANSACTION ISOLATION LEVEL

READ UNCOMMITTED;

START TRANSACTION;

SELECT Balance

FROM Account

WHERE Account_No = 10001;

Stored Procedures
Functions
bankdb
Tables
Views
Stored Procedures
Functions
bookflow_db
kludb
sys

administration Schemas
information:.....
Schema:
2520080069_ananya

```

205 • SET SESSION TRANSACTION ISOLATION LEVEL
206 READ UNCOMMITTED;
207 • START TRANSACTION;
208 • SELECT Balance
209 FROM Account
210 WHERE Account_No = 10001;
211

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Balance
65000.00

Account 18 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 64	13:55:46	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
✓ 65	13:55:46	COMMIT	0 row(s) affected
✓ 66	13:56:23	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 67	13:56:23	START TRANSACTION	0 row(s) affected
✓ 68	13:56:23	UPDATE Account SET Balance = Balance + 20000 ...	1 row(s) affected Rows matche
✓ 69	13:56:56	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 70	13:56:56	START TRANSACTION	0 row(s) affected
✓ 71	13:56:56	SELECT Balance FROM Account WHERE Account...	1 row(s) returned

Object Info Session

SESSION 1: ROLLBACK;

```

212 • ROLLBACK;
213

```

Output

Action Output

#	Time	Action	Message
✓ 65	13:55:46	COMMIT	0 row(s) affected
✓ 66	13:56:23	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 67	13:56:23	START TRANSACTION	0 row(s) affected
✓ 68	13:56:23	UPDATE Account SET Balance = Balance + 20000 ...	1 row(s) affected Rows matche
✓ 69	13:56:56	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 70	13:56:56	START TRANSACTION	0 row(s) affected
✓ 71	13:56:56	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
✓ 72	13:57:21	ROLLBACK	0 row(s) affected

SESSION 2:

SELECT Balance

FROM Account

WHERE Account_No = 10001;

COMMIT;

The screenshot displays a database management tool interface. At the top, a SQL query is entered in a text area:

```
214 • SELECT Balance
215 FROM Account
216 WHERE Account_No = 10001;
217 • COMMIT;
218
```

Below the query, the 'Result Grid' is visible, showing a single row of data:

Balance
65000.00

On the right side of the interface, there are icons for 'Result Grid', 'Form Editor', and 'Field Types'. Below these, a tab labeled 'Account 19' is active, and a 'Read Only' status is indicated.

The 'Output' section at the bottom shows a table of 'Action Output' with columns for '#', 'Time', 'Action', and 'Message'.

#	Time	Action	Message
67	13:56:23	START TRANSACTION	0 row(s) affected
68	13:56:23	UPDATE Account SET Balance = Balance + 20000 ...	1 row(s) affected Rows matched:
69	13:56:56	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
70	13:56:56	START TRANSACTION	0 row(s) affected
71	13:56:56	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
72	13:57:21	ROLLBACK	0 row(s) affected
73	13:57:42	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
74	13:57:42	COMMIT	0 row(s) affected

PART I – READ COMMITTED EXPERIMENT

SESSION 1:

SET SESSION TRANSACTION ISOLATION LEVEL

READ COMMITTED;

START TRANSACTION;

UPDATE Account

SET Balance = Balance + 10000

WHERE Account_No = 10001;

The screenshot displays a SQL IDE interface. On the left, a sidebar shows the 'Schema: 2520080069_ananya'. The main editor contains a SQL script with line numbers 219 to 226. The script is as follows:

```
219 SET SESSION TRANSACTION ISOLATION LEVEL
220 READ COMMITTED;
221 START TRANSACTION;
222 UPDATE Account
223 SET Balance = Balance + 10000
224 WHERE Account_No = 10001;
225
226
```

Below the script, the 'Output' pane shows the 'Action Output' table. It contains the following data:

#	Time	Action	Message
70	13:56:56	START TRANSACTION	0 row(s) affected
71	13:56:56	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
72	13:57:21	ROLLBACK	0 row(s) affected
73	13:57:42	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
74	13:57:42	COMMIT	0 row(s) affected
75	14:01:41	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
76	14:01:41	START TRANSACTION	0 row(s) affected
77	14:01:41	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa..

SESSION 2:

SET SESSION TRANSACTION ISOLATION LEVEL

READ COMMITTED;

START TRANSACTION;

SELECT Balance

FROM Account

WHERE Account_No = 10001;

226 • SET SESSION TRANSACTION ISOLATION LEVEL
 227 READ COMMITTED;
 228 • START TRANSACTION;
 229 • SELECT Balance
 230 FROM Account
 231 WHERE Account_No = 10001;
 232

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Result Grid

Balance
75000.00

Account 20 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 73	13:57:42	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
✓ 74	13:57:42	COMMIT	0 row(s) affected
✓ 75	14:01:41	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 76	14:01:41	START TRANSACTION	0 row(s) affected
✓ 77	14:01:41	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1
✓ 78	14:02:05	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 79	14:02:05	START TRANSACTION	0 row(s) affected
✓ 80	14:02:05	SELECT Balance FROM Account WHERE Account...	1 row(s) returned

SESSION 1:-

COMMIT;

233 • COMMIT;

Output

Action Output

#	Time	Action	Message
✓ 74	13:57:42	COMMIT	0 row(s) affected
✓ 75	14:01:41	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 76	14:01:41	START TRANSACTION	0 row(s) affected
✓ 77	14:01:41	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1
✓ 78	14:02:05	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 79	14:02:05	START TRANSACTION	0 row(s) affected
✓ 80	14:02:05	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
✓ 81	14:02:36	COMMIT	0 row(s) affected

SESSION 2:

SELECT Balance

FROM Account

WHERE Account_No = 10001;

COMMIT;

The screenshot displays a database management tool interface. At the top, a SQL query is entered in a text area:

```
235 • SELECT Balance
236 FROM Account
237 WHERE Account_No = 10001;
238 • COMMIT;
239
```

Below the query editor, a "Result Grid" tab is active, showing a single row of data:

Balance
75000.00

To the right of the result grid are icons for "Filter Rows:", "Export:", and "Wrap Cell Content:". Below these are icons for "Result Grid", "Form Editor", and "Field Types".

At the bottom, a tab labeled "Account 21" is open, showing a "Read Only" status. Below the tab is an "Output" section with a dropdown menu set to "Action Output". The output log contains the following entries:

#	Time	Action	Message
✓ 76	14:01:41	START TRANSACTION	0 row(s) affected
✓ 77	14:01:41	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 C
✓ 78	14:02:05	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 79	14:02:05	START TRANSACTION	0 row(s) affected
✓ 80	14:02:05	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
✓ 81	14:02:36	COMMIT	0 row(s) affected
✓ 82	14:03:49	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
✓ 83	14:03:49	COMMIT	0 row(s) affected

PART J – REPEATABLE READ EXPERIMENT

SESSION 1:

SET SESSION TRANSACTION ISOLATION LEVEL

REPEATABLE READ;

START TRANSACTION;

SELECT Balance

FROM Account

WHERE Account_No = 10001;

The screenshot displays a database management interface. At the top, a SQL script is shown with line numbers 240 through 246. The script contains the following commands:

```
240 • SET SESSION TRANSACTION ISOLATION LEVEL
241 • REPEATABLE READ;
242 • START TRANSACTION;
243 • SELECT Balance
244 • FROM Account
245 • WHERE Account_No = 10001;
246
```

Below the script, the 'Result Grid' is visible, showing a single row with the value 75000.00 for the column 'Balance'. The interface includes buttons for 'Filter Rows', 'Export', 'Wrap Cell Content', 'Result Grid', 'Form Editor', and 'Field Types'.

At the bottom, the 'Output' pane shows a table of actions and messages:

#	Time	Action	Message
79	14:02:05	START TRANSACTION	0 row(s) affected
80	14:02:05	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
81	14:02:36	COMMIT	0 row(s) affected
82	14:03:49	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
83	14:03:49	COMMIT	0 row(s) affected
84	14:04:22	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
85	14:04:22	START TRANSACTION	0 row(s) affected
86	14:04:22	SELECT Balance FROM Account WHERE Account...	1 row(s) returned

SESSION 2:

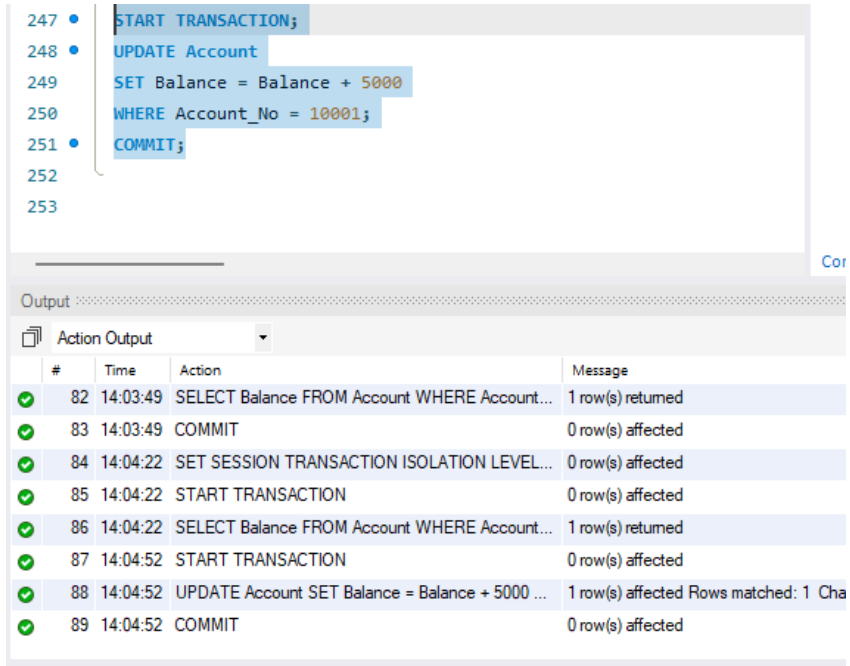
START TRANSACTION;

UPDATE Account

SET Balance = Balance + 5000

WHERE Account_No = 10001;

COMMIT;



The screenshot displays a SQL script editor with the following code:

```
247 • START TRANSACTION;  
248 • UPDATE Account  
249 SET Balance = Balance + 5000  
250 WHERE Account_No = 10001;  
251 • COMMIT;  
252  
253
```

Below the script editor is the 'Output' window, which shows the 'Action Output' table:

#	Time	Action	Message
✓ 82	14:03:49	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
✓ 83	14:03:49	COMMIT	0 row(s) affected
✓ 84	14:04:22	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 85	14:04:22	START TRANSACTION	0 row(s) affected
✓ 86	14:04:22	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
✓ 87	14:04:52	START TRANSACTION	0 row(s) affected
✓ 88	14:04:52	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Cha
✓ 89	14:04:52	COMMIT	0 row(s) affected

SESSION 1:

SELECT Balance

FROM Account

WHERE Account_No = 10001;

COMMIT;

```
253 • SELECT Balance
254 FROM Account
255 WHERE Account_No = 10001;
256 • COMMIT;
257
```

Result Grid	
	Filter Rows:
	Export:  Wrap Cell Content: 
Balance	
▶ 80000.00	

Result Grid

Form Editor

Field Types

Account 23 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 84	14:04:22	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
✓ 85	14:04:22	START TRANSACTION	0 row(s) affected
✓ 86	14:04:22	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
✓ 87	14:04:52	START TRANSACTION	0 row(s) affected
✓ 88	14:04:52	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 C
✓ 89	14:04:52	COMMIT	0 row(s) affected
✓ 90	14:05:11	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
✓ 91	14:05:12	COMMIT	0 row(s) affected

PART K – SERIALIZABLE EXPERIMENT

SESSION 1:

SET SESSION TRANSACTION ISOLATION LEVEL
SERIALIZABLE;
START TRANSACTION;
SELECT *
FROM Account
WHERE Account_No = 10001;

The screenshot displays a database management interface. At the top, a SQL query is entered in a text area:

```
258 • SET SESSION TRANSACTION ISOLATION LEVEL
259   SERIALIZABLE;
260 • START TRANSACTION;
261 • SELECT *
262   FROM Account
263   WHERE Account_No = 10001;
264
```

Below the query, a "Result Grid" shows the execution results. The grid has columns: Account_No, Customer_ID, Account_Type, Balance, and Branch. The first row shows the account details for Account_No 10001. A second row with all NULL values is also present.

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	80000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Below the result grid, an "Output" section shows a log of actions performed during the session:

#	Time	Action	Message
87	14:04:52	START TRANSACTION	0 row(s) affected
88	14:04:52	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1
89	14:04:52	COMMIT	0 row(s) affected
90	14:05:11	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
91	14:05:12	COMMIT	0 row(s) affected
92	14:05:41	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
93	14:05:41	START TRANSACTION	0 row(s) affected
94	14:05:41	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned

SESSION 2:

SET SESSION TRANSACTION ISOLATION LEVEL

SERIALIZABLE;

START TRANSACTION;

UPDATE Account

SET Balance = Balance + 5000

WHERE Account_No = 10001;

```
265 • SET SESSION TRANSACTION ISOLATION LEVEL
266     SERIALIZABLE;
267 • START TRANSACTION;
268 • UPDATE Account
269     SET Balance = Balance + 5000
270     WHERE Account_No = 10001;
271
```

Context

Output

Action Output

#	Time	Action	Message
90	14:05:11	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
91	14:05:12	COMMIT	0 row(s) affected
92	14:05:41	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
93	14:05:41	START TRANSACTION	0 row(s) affected
94	14:05:41	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
95	14:06:18	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
96	14:06:18	START TRANSACTION	0 row(s) affected
97	14:06:18	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Changed

SESSION 1:

COMMIT;

Context

Output

Action Output

#	Time	Action	Message
91	14:05:12	COMMIT	0 row(s) affected
92	14:05:41	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
93	14:05:41	START TRANSACTION	0 row(s) affected
94	14:05:41	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
95	14:06:18	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
96	14:06:18	START TRANSACTION	0 row(s) affected
97	14:06:18	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Change
98	14:06:48	COMMIT	0 row(s) affected

PART M – BANK TRANSACTION EXAMPLES

48. DEPOSIT TRANSACTION

START TRANSACTION;
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;
COMMIT;

Schema:
2520080069_ananya

The screenshot shows a SQL execution window with a list of statements on the left and an 'Output' pane on the right. The statements are: START TRANSACTION; (line 274), UPDATE Account (line 275), SET Balance = Balance + 5000 (line 276), WHERE Account_No = 10001; (line 277), and COMMIT; (line 278). The 'Output' pane shows a table of execution results with columns: #, Time, Action, and Message. The results show a sequence of operations including a SELECT statement, setting the transaction isolation level, and two successful UPDATE operations, each affecting 1 row.

#	Time	Action	Message
94	14:05:41	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
95	14:06:18	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
96	14:06:18	START TRANSACTION	0 row(s) affected
97	14:06:18	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1
98	14:06:48	COMMIT	0 row(s) affected
99	14:07:51	START TRANSACTION	0 row(s) affected
100	14:07:51	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1
101	14:07:51	COMMIT	0 row(s) affected

49. WITHDRAWAL TRANSACTION

START TRANSACTION;
UPDATE Account
SET Balance = Balance - 5000
WHERE Account_No = 10001;
COMMIT;

```

280 • START TRANSACTION;
281 • UPDATE Account
282   SET Balance = Balance - 5000
283   WHERE Account_No = 10001;
284 • COMMIT;
285

```

Context Help

Output

#	Time	Action	Message
✓ 97	14:06:18	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Changed: 1 W
✓ 98	14:06:48	COMMIT	0 row(s) affected
✓ 99	14:07:51	START TRANSACTION	0 row(s) affected
✓ 100	14:07:51	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Changed: 1 W
✓ 101	14:07:51	COMMIT	0 row(s) affected
✓ 102	14:08:10	START TRANSACTION	0 row(s) affected
✓ 103	14:08:10	UPDATE Account SET Balance = Balance - 5000 ...	1 row(s) affected Rows matched: 1 Changed: 1 W
✓ 104	14:08:10	COMMIT	0 row(s) affected

50. CANCEL WITHDRAWAL

```

START TRANSACTION;
UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001;

```

```

ROLLBACK;

```

Schema:
2520080069_ananya

```

286 • START TRANSACTION;
287 • UPDATE Account
288   SET Balance = Balance - 10000
289   WHERE Account_No = 10001;
290
291 • ROLLBACK;
292

```

Cont

Output

#	Time	Action	Message
✓ 100	14:07:51	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Char
✓ 101	14:07:51	COMMIT	0 row(s) affected
✓ 102	14:08:10	START TRANSACTION	0 row(s) affected
✓ 103	14:08:10	UPDATE Account SET Balance = Balance - 5000 ...	1 row(s) affected Rows matched: 1 Char
✓ 104	14:08:10	COMMIT	0 row(s) affected
✓ 105	14:08:37	START TRANSACTION	0 row(s) affected
✓ 106	14:08:37	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Char
✓ 107	14:08:37	ROLLBACK	0 row(s) affected

51. TRANSFER WITH SAVEPOINT

```
START TRANSACTION;
UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001;
SAVEPOINT AfterDebit;
UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10002;
COMMIT;
```

The screenshot shows a database management interface. On the left, there is a sidebar with 'Administration' and 'Schemas' tabs. Under 'Schemas', a schema named '2520080069_ananya' is selected. The main area displays a SQL script with line numbers 292 to 302. The script is as follows:

```
292
293 • START TRANSACTION;
294 • UPDATE Account
295   SET Balance = Balance - 10000
296   WHERE Account_No = 10001;
297 • SAVEPOINT AfterDebit;
298 • UPDATE Account
299   SET Balance = Balance + 10000
300   WHERE Account_No = 10002;
301 • COMMIT;
302
```

Below the script, there is an 'Output' section with a dropdown menu set to 'Action Output'. It displays a table of execution results:

#	Time	Action	Message	D
✓ 105	14:08:37	START TRANSACTION	0 row(s) affected	0.0
✓ 106	14:08:37	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...	0.0
✓ 107	14:08:37	ROLLBACK	0 row(s) affected	0.0
✓ 108	14:08:58	START TRANSACTION	0 row(s) affected	0.0
✓ 109	14:08:58	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...	0.0
✓ 110	14:08:58	SAVEPOINT AfterDebit	0 row(s) affected	0.0
✓ 111	14:08:58	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Changed: 1 Wa...	0.0
✓ 112	14:08:58	COMMIT	0 row(s) affected	0.0

At the bottom left, there are tabs for 'Object Info' and 'Session'.